Dustin Biser

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Interests and Focus

Real-Time Fluid Simulation, GPU Computing, and Computer Graphics.

Education

University of Waterloo
Master of Mathematics, Computer Science 2014 – present

University of Maryland, College Park
Bachelor of Science, Physics 2012
Bachelor of Science, Mathematics 2012

Current Research

Computational Motion Group
Sept. 2014 – Present
University of Waterloo
Advisor: Dr. Christopher Batty
Physics-based animation with an emphasis on real-time fluid simulation and GPU computing.

Recent Projects

Fluid Simulation Link
Sept 2014 – Present
Grid-based and particle-based methods for real-time fluid simulation.

3D Graphics Engine Link
July 2013 – Present
Engine for interactive 3D graphics. Developed using C++, OpenGL 4.0+. Capabilities include: mesh loading, shader management, camera system, and real-time lighting effects.

Employment

Computer Graphics Software Engineer (R&D)
January 2014 – August 2014 (8 months)
Nemetschek Vectorworks (Columbia, MD)
Architectural CAD company that develops 3D modeling software.

- C++, OpenGL development of in-house graphics engine used to facilitate seamless real-time walkthroughs of architectural sites for OSX, Windows, and iOS devices.
- Implemented several subsystems that included:
    - Camera system,
    - Texture mapping system,
    - Transparency effects,
    - Skyboxes, and
    - Intuitive user input controls for traversing various environments.

Employment (continued)

Computer Vision Software Engineer
June 2012 – March 2013 (10 months)
SAIC (Arlington, VA)
Government contractor focused on a mix of research and building sophisticated software applications for the military and the Department of Defense.

- C++, OpenGL, Qt development of 3D terrain rendering application used for military exercise planning and training.
- Java, OpenGL development of multithreaded application used for stabilizing and stitching together video frames from unmanned aerial vehicles in real time.

Prior Research Experience

Undergraduate Research Intern
May 2009 – August 2009 (4 months)
National Institute of Standards and Technology (Gaithersburg, MD)
Developed an interactive 3D graphical solution for interpolating and displaying the amounts of radiation delivered throughout containers that had been sent through industrial irradiation facilities.

Undergraduate Research Intern
May 2008 – August 2008 (4 months)
National Institute of Standards and Technology (Gaithersburg, MD)
Created and validated a software controller for programming electro-optical devices used in high-fidelity remote sensing instrument calibrations.

Main Skills

★ Very strong communication skills.
★ Very strong knowledge of 3D graphics theory.
★ Strong knowledge of fluid and rigid body dynamics.
★ Strong math and physics background.
★ Fluent in modern C++, GLSL, and Java.
★ Experienced in GPU computing, real-time lighting, and simulation.

Additional Skills

★ Libraries: OpenGL, STL, GLM, GLFW, Boost, Qt, GoogleTest.
★ Art and Modeling: Blender, Gimp, Photoshop.
★ Dev. Tools: XCode, AppCode, Visual Studio, Eclipse, Git, Perforce, Premake4, Cmake.
★ Operating Systems: OSX, iOS, Ubuntu, Windows.